Retinal Disease Detection and Classification Using Convolution Neural Networks and Transfer Learning

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Abstract. The categorization of retinal diseases utilizing Optical Coherence Tomography (OCT) images has garnered considerable interest in the domain of computational medical imaging. This research paper introduces a deep learning methodology for the automated classification of OCT images via a custom Convolution Neural Network (CNN) in conjunction with recognized transfer learning models. Deep learning methodologies are favored for their exceptional accuracy and performance; however, they frequently necessitate substantial computational resources and time for training. MobileNet, ResNet50, GoogleNet, and DenseNet are employed for feature extraction and comparison analysis. The models are trained and refined on an OCT dataset utilizing various hyperparameter settings, such as learning rate, epoch count, and optimizing techniques to get maximal accuracy. Upon concluding our study, the performance of these models is evaluated, and the optimal design is ascertained based on accuracy and additional assessment criteria. Our methodology illustrates the capability of deep learning in the automated classification of OCT images, providing enhanced diagnostic support for the identification of retinal diseases

Keywords: OCT images, CNN, transfer learning, deep learning, feature extraction

1 Introduction

There have been heights of progress in medical imaging from diagnosis all the way through to treatment, and an important role is played by Optical Coherence Tomography (OCT) in ophthalmology. Optical Coherence Tomography (OCT) provides high-resolution cross-sectional imaging of the retina in the diagnosis of Age-related Macular Degeneration (AMD), Diabetic Retinopathy (DR), and Glaucoma. It is only by rapid identification and accurate classification that the medical treatments may be implemented immediately, which would otherwise save these patients from a significant impairment in their vision. Conventionally, manual diagnosis has been the bedrock of diagnostics. This is a very time-effective procedure but inherently biased because it relies so much on the skill of the operator, which makes it even more vulnerable to human errors. The automation of medical imaging classification, facilitated by the advent of deep learning (DL), has resulted in the development of enhanced tools and methodologies that enable the achievement of these objectives with greater efficiency.

Deep learning, especially Convolutional Neural Networks (CNNs), has brought about the automation of feature extraction and acquisition of hierarchical representations of medical images so fast with its introduction that it can be said to have redefined the landscape of OCT image categorization. Unlike traditional methods, CNNs do not require manual construction of features; rather, the model learns to recognize patterns and abnormalities directly from raw image data. Training deep networks from scratch requires a huge amount of labelled data and a tremendous amount of computing power; thus, it is an expensive affair. To tackle these issues, transfer learning has been favoured, i.e., fine-tuning models pre-trained on huge datasets (like ImageNet) for medical imaging applications. Transfer learning therefore not only minimizes training time but also enhances accuracy and thus becomes an attractive option for OCT classification.

An assessment and comparative analysis will be performed to evaluate the effectiveness of deep-learning models for the categorization of OCT images, thereby providing insight into the practicality of CNN-based approaches in ocular disease detection. By quantifying different architectures and optimizing their hyperparameters, this study would contribute to the design of more efficient, reliable, and automated diagnostic tools for retinal disease classification.

Injecting new data into the system increases categorization accuracy. This would, however, mean working with large datasets with heavy demands on computing resources and processing time for trustworthy accuracy. Transfer functions were used in this research to solve the stated problem, whereby a model is enabled to use properly trained neural network knowledge by making minor adjustments mainly to the weights and biases of the neurons toward a new dataset. Transfer learning enables the knowledge-sharing process among deep learning models. The transfer learning with pre-trained architectures such as MobileNet, ResNet50, GoogleNet, and DenseNet provides better performance with efficient training, making it a viable and productive method for OCT image classification.

2 Context

The optical coherence tomography method is a diversified process widely used by oculists to give a high resolution of the cross-sectional captures of the membrane. This non-invasive technique provides accurate visualization of the retinal layers to diagnose and track various retinal disorders. There are benefits to early detection. This age disease is a degenerative disease that can serve to severely impair sight if not detected and treated early. It is considered one of the important problems in the field of ophthalmology.

The area of medical imaging and artificial intelligence has expanded exponentially, especially in the case of ophthalmology, with deep learning models being used more and more for disease classification. Optical Coherence Tomography (OCT) is a very effective imaging modality for the assessment of the retina, yielding high-resolution cross-sectional images of the retina to assist in the disease diagnosis of conditions such as age-related macular degeneration (AMD), diabetic retinopathy (DR), and other retinal disorders [4]. Accurate classification of such conditions is crucial in early diagnosis and effective medical treatment. Conventional diagnosis relies on expert interpretation, which is subjective, time-consuming, and prone to inter-observer variability. The use of deep learning models for computer-aided OCT image classification has been proposed as a feasible alternative to improve diagnosis accuracy and effectiveness.

Convolutional Neural Networks (CNNs) have been shown to have remarkable success in image classification because CNNs can learn and extract hierarchically abstract features from images automatically. Training a CNN from scratch involves a huge amount of labeled data and heavy computer resources. In clinical use, it is difficult to obtain large, well-annotated datasets because professional annotation and heavy time consumption are required. Moreover, training deep networks on these datasets is a computationally intensive task, which consumes heavy computer resources. Transfer learning has been extensively used in medical imaging tasks in an attempt to counter these issues.

The best transfer learning is combined with different deep learning models, such as MobileNet, ResNet50, GoogleNet, and DenseNet [2-3]. The models have been categorized into four types: Normal, Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen are shown in Fig. 1.

Normal. Retinal images exhibit health with no indications of abnormalities or disease.
Drusen. Small, yellowish buildups of dead cells that happen below the retina are seen as the first sign of age-related macular degeneration (AMD), which can lead to worsening vision over time.
Choroidal Neovascularization. A disorder characterized by the excessive growth of blood vessels behind the retina, mostly leading to visual impairment
Diabetic Macular Edema. It is a severe complication of diabetic retinopathy that is characterized by the deposition of fluid in the macula, resulting in vision impairment. This enlargement in central vision is caused by the leakage of blood vessels in the retina.

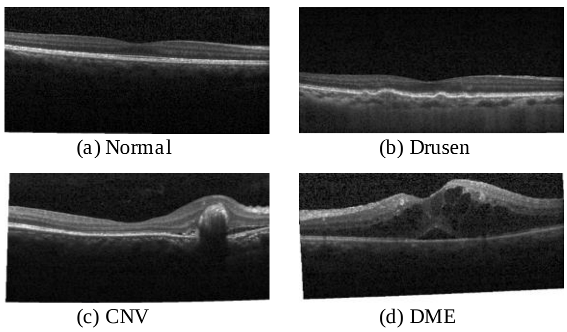


Fig. 1. Sample images OCT image dataset

3 Methodology

The research workflow uses a structured method to divide OCT images into four groups: Normal, Choroidal Neovascularization (CNV), Drusen, and Diabetic Macular Edema (DME). The method starts with the aggregation of the dataset, which is partitioned into training, testing, and validation subsets. The images are subsequently preprocessed to improve model accuracy. This includes resizing, enlarging, and augmenting techniques like flipping, rotating, and changing the contrast to make the model more stable and less likely to overfit.

This study uses a systematic strategy in the building and evaluation of deep learning models for the classification of retinal images. This is followed by two classes: binary and multiclass classifications, aiming to enhance classification accuracy. The binary classification identifies three subsets: testing, training, and validation. These subsets consist of four additional sets: normal, DME, CNV, and drusen. On the other hand, the multiclass classification uses the same method as the binary classification. This divide made it possible to explore the performance of different models in greater detail and with better understanding.

The flowchart below details the workflow of our study, includes all the steps were shown in Fig. 2.

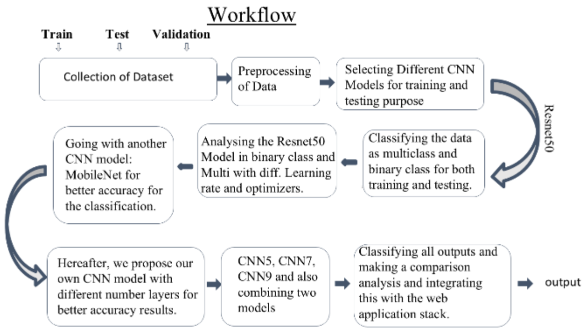


Fig. 2. Workflow Representation

The models were trained and evaluated after conducting a comprehensive investigation into retinal image categorization. All the images across each category of the dataset are evenly distributed so that the divisions for training, validation, and testing are perfectly fair and dependable in the outputs. The optimization of hyperparameters, ranging from learning rate, optimizers, and even batch sizes, was considered to understand the best-suited one for a particular model.

Below is the description of the models used in this project.

3.1 ResNet50

The ResNet50 model is one of the most commonly used convolutional neural networks with 50 layers, chosen for its ability to overcome gradient problems through residual connections. The model was pre-trained on ImageNet and then fine-tuned on the dataset.

Binary and multiclass classification tasks were performed by freezing the early layers to retain generic extraction capability while training the fully linked layers on our data. Different optimizers like SGD, ADAM, NDAM, and ADAMAX were used for getting different accuracies, and it was trained at different learning rates that resulted in maximum accuracy. With an overall accuracy of 96.00%, this model ranks among the top-performing models in our society [5-6].

3.2 DenseNet

The DenseNet method made it easier to reuse features and reduce the number of parameters by tightly connecting the design so that each layer gets inputs from all pairs. Similar to the ResNet50 model, this model also uses different optimizers with different learning rates for better accuracy. For our study, the DenseNet model came out to be the best model with an overall accuracy of 98.00%.

3.3 GoogleNet

. This model provides us with a high level of accuracy for our study. GoogleNet, also known as Inception v1, employs a multilevel system that enables the model to gather features at various scales. The model was trained and tested on the dataset, with the first layer fixed to facilitate pre-learning information. Various combinations of optimizers and learning rates were also experimented with to enhance the model's performance. GoogleNet showed the best performance, and it was a reliable choice for retinal image classification.

3.4 Custom CNN Model

A custom CNN model was implemented in our study, which can be stated as our own designed model with a layer of five, which gives a good amount of accuracy, and implemented and compared its performance with the pre-trained models. The architecture of CNN consists of layers of convolution followed by pooling layers and the activation function [1], taken as ‘Leaky-relu,’ in addition to batch normalization. The model was trained from the beginning using the dataset that was defined; careful tuning of the hyperparameter algorithms was used, including the learning rate and batch sizes, to achieve the best accuracy.

Each model went through tough training, testing, and evaluation using both binary and multiclass classification methods. The results were analyzed to identify the most effective architecture in terms of accuracy, loss, and efficiency. A thorough comparison was also performed by integrating the different model results to determine the most appropriate model.

Hardware Detailing. The performance of a first-version prototype is based on the hardware requirements and results in reduced execution time and enhanced processing capability. This, in turn, enhances the overall performance. The hardware used in the prototype model for deploying the web application is mentioned here. The neural network model was developed and implemented on a system that met the following specifications.

The hardware description of the system specification is listed below in Table 1.

Table 1. Hardware Detailing's

Sl. No.	Specifications	
	Name	Details
1.	Processor	Intel Core i5-1240P Processor
2.	RAM	16 GB 1.70GHz
3.	GPU	Intel iris Xe (2GB)

Software detailing. PyTorch is a Python-based framework specifically designed for neural network computation, which was employed to develop and implement the neural network models pertaining to this research. Its endless connectivity with Python modules like NumPy, SciPy, and Pillow renders it a highly versatile and efficient option for the model.

This work used pre-trained deep learning models such as ResNet50, GoogleNet, DenseNet, and custom CNN models. Their performance was improved by applying transfer learning to the dataset.

TorchVision was utilized for image transformations and data loading, Pillow for complex image preprocessing, and Flask to develop the web application framework, in addition to PyTorch. This integration enabled us to create and build a resilient system for real-time classification and analysis for efficiency and simplicity of execution.

Web Application Methodology. The web application was developed to offer interactive and user-friendly autonomous behavior with the users for classifying retinal images using the DenseNet model because this model is our base exhibited from the accuracy point of view in the training phase. The web application was developed with Flask and a lightweight Python framework to handle the server. Flask was selected for its simplicity, durability, and interoperability with other Python packages. The principal backend functions encompass routing user requests, loading the pre-trained DenseNet model, handling the performance report when the user is interacting with the website, and managing the classification pipeline.

The Python served as the primary machine and deep learning framework for loading the datasets. The TorchVision library used for utilizing the data augmentation and preprocessing operations includes scaling and normalization. Furthermore, the Pillow (PIL) library was employed to manage picture uploads and convert them from the user to the input of the model format. The user-side interface was developed using HTML, CSS, and JavaScript to enable the user to upload the retinal scans and examine the classification results. After the image submission, the backend processes the images, utilizes the DenseNet model, and returns the projected class along with a confidence score.

The model was made optimized with minimal computational errors, thereby enhancing the application's performance and ensuring smooth functionality on systems with the hardware resources. Various optimizers and learning rates have been chosen to get the best accuracy results and up to a degree of efficiency.

The online application underwent comprehensive testing to verify its compatibility and durability in classifying retinal pictures and give a classification report. The integration of Flask, PyTorch, torchvision, and Pillow gave a better development and a high-performance output that connects advanced deep learning models with practical usability.

4 Results and Discussions

The result of this study depicts the diverse models and their approaches used to categorize Optical Coherence Tomography (OCT) images. Hereby the dataset was divided into training, testing, and validation micro sets to provide balanced results and check overfitting. A series of tests were being performed, altering the model architecture and changing the optimizer and learning rates to determine the most optimal results.

The dataset was initially preprocessed and classified into binary categories (normal versus AMD) and multiclass categories (normal, CNV, DME, and drusen). Binary classification consistently attained higher accuracy owing to its simplicity, although multiclass classification provided substantial insights into the dataset's complexity. The models underwent training and testing using predetermined splits, and their performance metrics—accuracy, precision, recall, and F1 score—were recorded for thorough analysis.

The data was classified into two classes: binary and multiclass. The ResNet50 model was trained and tested with three different optimizers to achieve the best output, using varying learning rates and optimizers. Below is the comparison table along with the graph for each binary class for the ResNet50 model.

Firstly, the Adam optimizer is shown below in Table 2, 3 and 4 along with a bar plot in Fig. 6, and below the rest of the classes are displayed as follows:

Firstly, the Adam optimizer is shown below in Table 2, 3 and 4 along with a bar plot in Fig. 3, and below the rest of the classes are displayed as follows:

4.1 Adam Optimizer

Table 2. Comparison table of Normal-CNV

Learning rate	Class	Accuracy
0.0001	Normal - CNV	100%
0.001		93%
0.01		50%

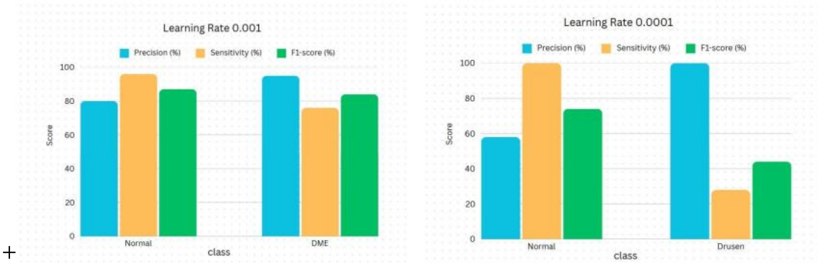
Table 3. Comparison table of Normal DME

Learning rate	Class	Accuracy
0.0001	Normal - DME	50%
0.001		86%
0.01		51%

Table 4. Comparison table of Normal-Drusen class

Learning rate	Class	Accuracy
0.0001	Normal - Drusen	64%
0.001		50%
0.01		50%

For this optimizer, the model's performance is evaluated using metrics such as precision, sensitivity, and F1 score. The following graphs provide a detailed comparison of each learning rate for the binary class.



(a) (b)

Fig. 3. Bar chart of Adam optimizer learning rate: (a) 0.001 and (b) 0.0001

Secondly, the Nadam optimizer is shown below in Table 5, 6, and 7, along with a bar plot in Fig. 4.

4.2 Nadam Optimizer

Table 5. Comparison table of Normal-CNV

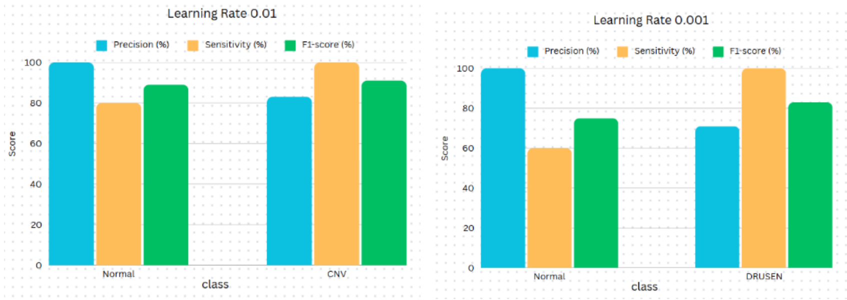
Learning rate	Class	Accuracy
0.0001	Normal - CNV	83%
0.001		49%
0.01		90%

Table 6. Comparison table of Normal DME

Learning rate	Class	Accuracy
0.0001	Normal - DME	98%
0.001		99%
0.01		45%

Table 7. Comparison table of Normal-Drusen class

Learning rate	Class	Accuracy
0.0001	Normal - Drusen	73%
0.001		80%
0.01		52%



(a) (b)

Fig. 4. Bar chart of Nadam optimizer learning rate: (a) 0.01 and (b) 0.001

4.3 Adamax Optimizer

Table 8. Comparison table of Normal-CNV

Learning rate	Class	Accuracy
0.0001	Normal - CNV	91%
0.001		50%
0.01		100%

Table 9. Comparison table of Normal DME

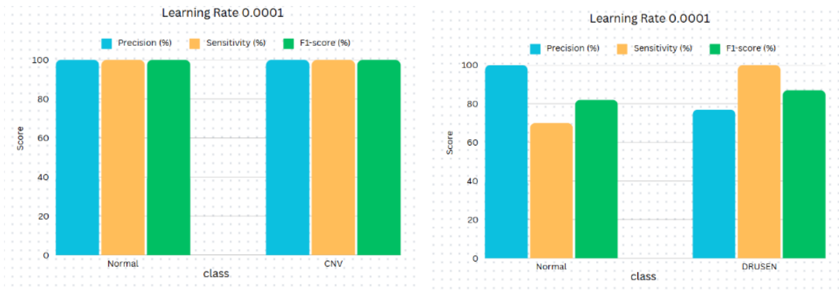
Learning rate	Class	Accuracy
0.0001	Normal - DME	51%
0.001		88%
0.01		51%

Table 10. Comparison table of Normal-Drusen class

Learning rate	Class	Accuracy
0.0001	Normal - Drusen	85%
0.001		63%
0.01		53%

Hereby, finally in the results section, a bar chart of Adamax optimizers with different learning rates and scores are shown in Table 8, 9 and 10 along with Fig. 5.

The parameters for the multiclass classification and detection using transfer learning and deep learning depend on various factors, which come from the confusion matrix. The model's performance also depends on the number of epochs used for training and testing, the learning rate applied in the study, and the type of optimizer employed.



(a) (b)

Fig. 5. Bar chart of Adamax optimizer learning rate: (a) 0.0001 and (b) 0.001

The outcome of the proposed study is assessed using the confusion matrix. Within the matrix, the quantities of True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN) yield accurate data findings, but also false positives (FP) and false negatives (FN) indicate misclassified results from the model. Thus, precision, F1-score, recall, and accuracy can be used to assess the machine's performance.

$$\text{Accuracy (A)} = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

The False Positive Rate (FPR) measures the frequency with which the model categorizes negatives and positives. An ideal classifier expects FPR to be 0. Denoted as:

$$\text{FPR} = \frac{FN}{FN+TP} \tag{2}$$

The true value, or the positive value, is given by the precision. It reflects the system's ability to avoid false positives.

$$\text{Precision} = \frac{TP}{TP+FP} \tag{3}$$

Recall is also sensitivity that gives information about how the model decreases FN; it can be measured as

$$\text{Recall} = \frac{TP}{TP+FN} \tag{4}$$

Here, the F1 score specifies the harmonic mean of precision and recall.

$$\text{F1-score} = \frac{2.TP}{2.TP+FN+FP} \tag{5}$$

The above graphs and comparison table all summarize the various scores predicted by the model with different optimizers and with the different learning rates.

Now the classification comparison table of four models is: ResNet50, GoogleNet, DenseNet, and Custom CNN, in which different parameters are shown like precision, F1-score, sensitivity, and accuracy were shown in Table 11, 12, 13 and 14. Below is the model summary along with the custom CNN model, which was our custom model with five layers and different activation functions. Fig. 6 displays the model comparison graph, illustrating the various models and their corresponding accuracies and losses.

Table 11. Comparison table of DenseNet model

Model	Precision	F1-Score	Recall / Sensitivity	Accuracy
Normal	98%	100%	99%	98%
DME	98%	98%	97%	
CNV	96%	98%	97%	
Drusen	100%	96%	98%	

Table 12. Comparison table of CNN Model

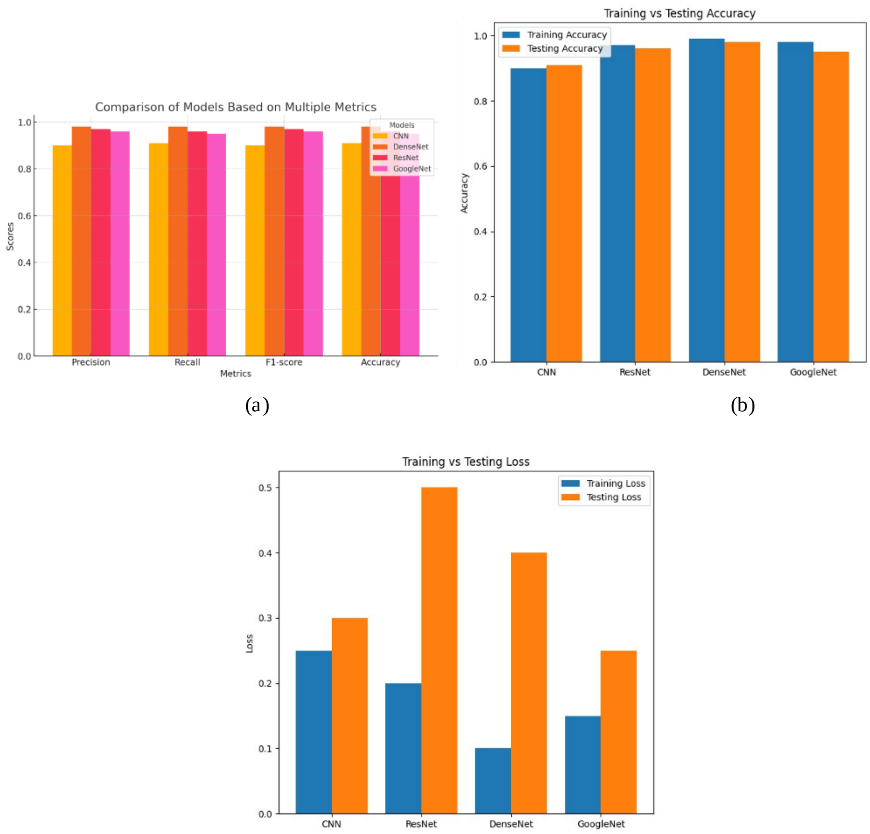
Model	Precision	F1-Score	Recall / Sensitivity	Accuracy
Normal	86%	86%	86%	91%
DME	90%	88%	89%	
CNV	90%	92%	91%	
Drusen	96%	96%	96%	

Table 13. Comparison table of RESNET50

Model	Precision	F1-Score	Recall / Sensitivity	Accuracy
Normal	100%	94%	97%	96%
DME	92%	98%	95%	
CNV	94%	98%	96%	
Drusen	100%	96%	98%	

Table 14. Comparison table of GoogleNET

Model	Precision	F1-Score	Recall / Sensitivity	Accuracy
Normal	96%	98%	97%	95%
DME	100%	96%	98%	
CNV	94%	94%	94%	
Drusen	92%	94%	93%	



(a) (b)

(c)

Fig. 6. Graphs depicting (a) Model comparison with different parameters, (b) Model accuracy comparison, and (c) Model loss comparison

Modern deep learning models like CNNs, DenseNets, ResNets, and GoogleNet have demonstrated exceptional performance in multiclass classification tasks. These models gather intricate information from high-dimensional OCT data, which allows for the precise differentiation between similar scenarios. Evaluation of these models' performances by metrics such as accuracy, precision, recall, and F1-score helps to determine the most successful strategy for clinical implementation.

The graph depicts how each model is classifying OCT images by showing that it performs well on all metrics. However, there are also some minor variations that a model outperforms. Here GoogleNet and ResNet outperform CNN and DenseNet in most categories. For this study, DenseNet was chosen as the fully functioning model, demonstrating excellent accuracy compared to the others.

DenseNet, due to excellent performance on most evaluation metrics—accuracy, precision, recall, and F1-score—will be the study's final model. It fits well with complicated tasks such as OCT image classification because of the architecture, promoting feature reuse while minimizing overfitting by cutting down the parameters. The accuracy graphs for training and testing show that the model is very good at generalization and can depend on unknown input, which shows that it has a constant learning curve. The confusion matrix also gives a detailed look at the results of the classification, showing how well DenseNet can put images into the normal, DME, CNV, and drusen groups. The ROC curve score in Fig. 7 highlights the model's ability to distinguish between positive and negative classes. All together, these confirm DenseNet's performance and dependability as shown in Fig. 8, making it the best model for our research and study and its use in imaging techniques. It is perfect for large-scale image datasets because it matches and combines the dense connection with the matching parameter utilization to give both useful classification performance and processing growth.

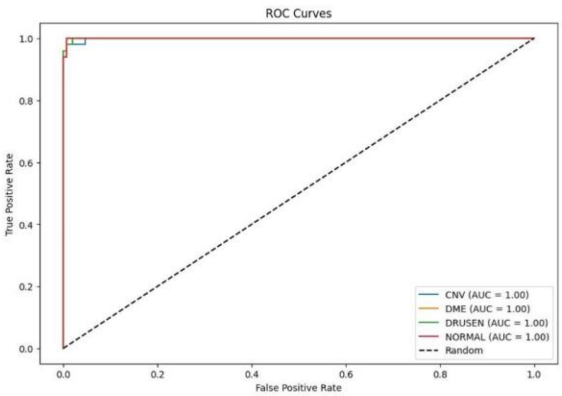
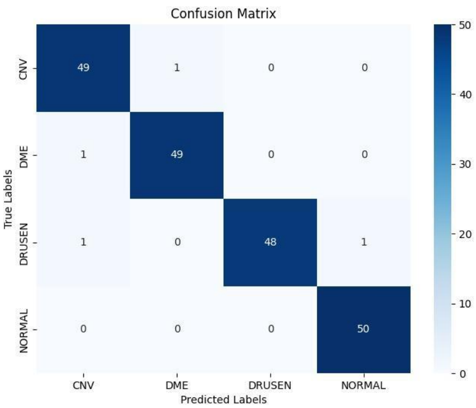
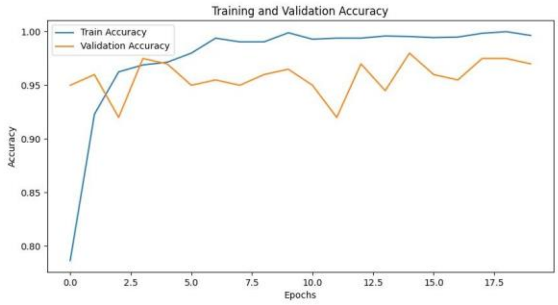


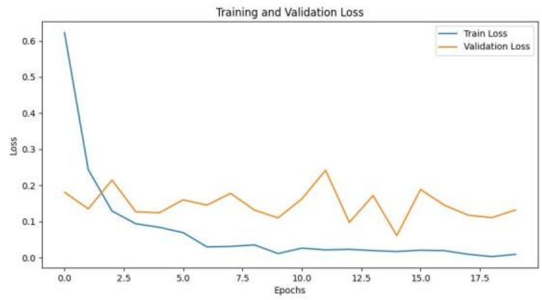
Fig. 7. DenseNet model ROC curve



(a)



(b)



(c)

Fig. 8. The above graphs depict the DenseNet model: (a) confusion matrix, (b) accuracy curve, and (c) loss curve

A website has now been designed with features that provide seamless functionality on both laptops and mobiles. While the mobile version is designed for small, responsive navigation, providing usage and accessibility for the laptop version, it provides a full, spacious design suitable for thorough exploration. The design effectively presents outputs across various environments and screen sizes, ensuring a consistent user experience.

5 Conclusion

The results of this study show that machine learning and deep learning models can effectively sort OCT images into separate groups based on conditions like Normal, Drusen, CNV, and DME. Among the models tested, DenseNet outperformed CNN and ResNet by 3-5% in precision, recall, and F1 score, achieving the highest accuracy of 98%. To enable practical implementation, an interactive website was developed that integrates DenseNet's output in a visually appealing and user-friendly format, bridging advanced AI technology with its application in medical imaging. This platform provides an intuitive interface for effective result analysis. Overall, the study not only highlights how AI models like DenseNet can improve diagnostic accuracy in imaging but also emphasizes the importance of user-centered design for real-world applications. Future research could focus on expanding the dataset to include more diverse conditions, refining the AI model, and drawing broader conclusions. This work lays a solid foundation for the use of AI in ophthalmology, contributing to the early detection and more effective treatment of retinal disorders.

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